Environmental Stress, Political Incentives, and Social Norms: Essays on the Determinants of Individual Behavior

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What Drives Individual Behavior?

- This thesis studies how different external factors shape individual behavior across three contexts
- **Chapter 1:** Temperature → Domestic violence
- **Chapter 2:** Lobbying → Politicians' trading
- **Chapter 3:** Siblings → Gender norms
- Across chapters, I combine rich data with quasi-experimental variation to identify causal effects

Chapter 1

Temperature Variations and Domestic Violence

Do Temperature Variations Influence Domestic Violence?

- 1 in 3 women worldwide have been subjected to physical and/or sexual violence
- Environmental stressors as one driver
- Research question:
  - Do Temperature Variations Influence Domestic Violence?
- Setting: Mexico City, a city with mild temperatures Plot

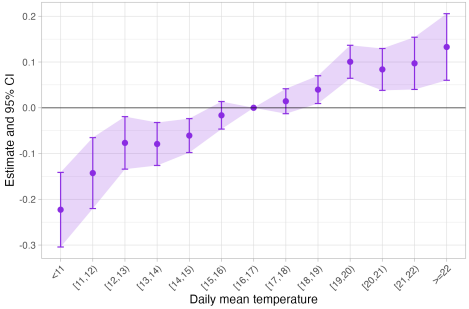
Data

- Weather data: hourly readings aggregated to the daily level
- Domestic violence data:
  - Police reports
  - Helpline calls
- Two key advantages:
  - High-frequency measurement
  - Independent validation source
- Approximately 2,400 neighborhoods over 1,500 days

Empirical Strategy

- Identification from within-neighborhood variation over time
- Conditioning on:
  - Rich set of temporal FE
  - Weather controls (precipitation, humidity, windspeed)
- Assumption: after conditioning, temperature is as good as random
- I estimate a flexible specification: Temperature bins

Main Result: Temperature - DV Relationship



- The relationship is approximately linear
- A 1°C increase in daily temperature leads to a **2.8% increase** in domestic violence reports
- The effect appears even at moderate temperatures

Linear results

Dynamic Adjustments and Cumulative Effects

- No evidence of displacement (harvesting)
- No evidence of persistence beyond same day
- Cumulative effects $\approx$ contemporaneous effect
- DV incidents induced by temperature are additional and contemporaneous

Heterogeneity by Neighborhood Income

- Effects are stronger in poorer neighborhoods
- Effects are stronger where housing quality is lower
- Interpretation:
  - Vulnerability depends on the ability to cope
- Implication:
  - Climate stress does not affect everyone equally — it amplifies existing inequalities

Robustness and Validity

- The result is replicated using helpline calls
- No meaningful effect on reporting timing
- The main result is not driven by pollution

→ The estimates reflect real behavioral responses

Takeaways

- Influence of temperature is immediate, linear, and unequal across neighborhoods
- Evidence points to short-run behavioral mechanisms
- Temperature acts as a continuous risk factor for domestic violence

Chapter 2

Trading on the Hill: How Lobbying Informs Congressional Stock Trading

Joint with Marco Battaglini, Valerio Leone Sciabolazza, and Eleonora Patacchini

Do Politicians Get Private Returns from Contacts with Lobbyists?

- Firms benefit from political connections
- Less is known about the private returns politicians may obtain from these relationships
- Research question:
  - Do legislators start trading in specific sectors after meeting with lobbyists from those sectors?
- Setting: U.S. Congress (House and Senate), 2011–2022

Data and Institutional Setting

- Lobbying Disclosure Act (1995):
  - Records meetings between lobbyists and elected officials
- STOCK Act (2012):
  - Requires members of Congress to disclose financial transactions within 45 days
- Meetings and stock trades are matched using a common sector classification
- Final dataset:
  - Panel at Legislator $\times$ sector $\times$ quarter

Empirical Strategy

- Treatment:
  - First lobbying exposure in a sector
- Outcome:
  - Whether the legislator trades in that sector in the following quarter
- Stacked difference-in-differences design
  - Because exposure is staggered across legislator-sector pairs
- Focus on legislators without prior trading in that sector

Identification Intuition

- Compare trading before and after first exposure to lobbying
- Control group:
  - Legislator–sector pairs not yet exposed in the same calendar period
- Sector-level shocks are absorbed by the design
- Interpretation:
  - Intent-to-treat effect of first lobbying exposure on subsequent trading

Main Result

- First-time lobbying exposure increases sector-specific trading by **25–40%**
- Robust across specifications and sector definitions
- Stronger for:
  - House members
  - Final-term legislators
- Suggests a role for institutional incentives and career horizon

Interpretation and Implications

- Evidence consistent with use of lobbying contacts as informational input
- Does not directly identify illicit trading
- Private returns to office may arise through access to information
- Implications for ongoing debates:
  - Transparency, Conflict-of-interest, Congressional trading bans
- The next big question is no longer just: "Do legislators start trading after lobbying exposure?" but: "Do they trade better once they become connected?"

Chapter 3

Random Sisters: Gender-Norm Formation in Early Adolescence

Joint with Libertad González

Do Siblings Shape Adolescents' Gender Norms?

- Gender norms matter for later educational and labor-market outcomes
- We do not fully understand how these norms are formed
- Research question:
  - Does having a sister, rather than a brother, affect adolescents' gender norms?
- Setting: adolescents aged 12–14 in Barcelona

Design and Measurement

- School-based lab-in-the-field study in Barcelona
- Rich direct measures of gender norms:
  - Gender attitudes
  - Perceived social norms
  - Incentivized coordination games
- Identification:
  - Conditional on having a second child; Sex of the second-born sibling is as-good-as-random

Main Result

- No clear detectable effect of sibling sex on aggregate indices of gender norms
- Null effects are broadly consistent across outcomes
- Main takeaway:
  - In early adolescence, sibling sex appears to play a limited role

Interpretation and Limits

- Parents, peers, and school environments may matter more than sibling sex
- Main limitation:
  - Modest statistical precision
- Contribution:
  - Credible evidence that sibling effects, if present, are limited at this stage of life

My thesis:

- Across three essays, I study how external conditions shape individual behavior
  - Temperature acts as a continuous risk factor for domestic violence, especially in vulnerable neighborhoods
  - Lobbying exposure is associated with greater sector-specific trading by legislators
  - Sibling sex has limited detectable effects on adolescents' gender norms in this setting

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Thank you!

Appendix

Mexico City: Mild Temperatures Year Round

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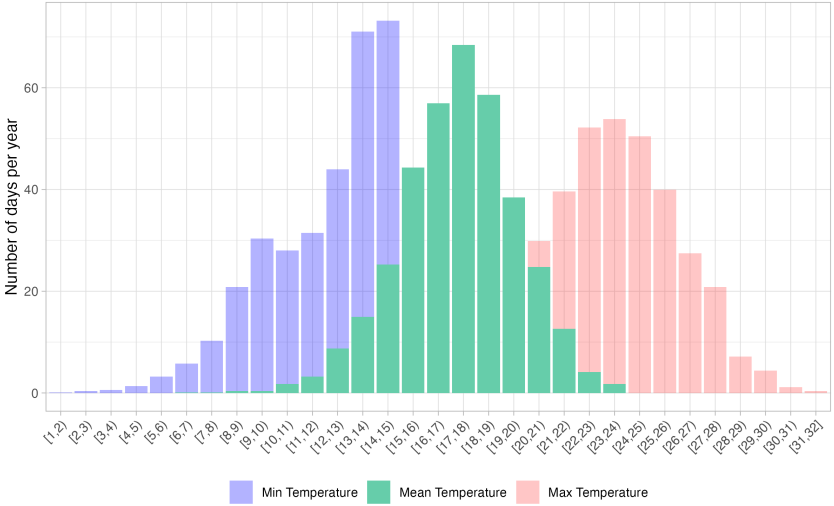

- Temperature = tmean = average of 24 hourly readings

Main Specification Results

Back

	Reports for DV		
	(i)	(ii)	(iii)
Panel A:			
24-hour Mean Temperature	0.0274***		
	(0.0034)		
Observations	3,624,826		
Panel B:			
Maximum temperature	0.0197***		0.0159***
	(0.0026)		(0.0030)
Minimum temperature		0.0178***	0.0097***
		(0.0029)	(0.0034)
Observations	3,624,826	3,624,826	3,624,826
Panel C:			
Daytime temperature	0.0246***		0.0220***
	(0.0030)		(0.0041)
Nighttime temperature		0.0188***	0.0039
		(0.0031)	(0.0042)
Observations	3,624,815	3,624,809	3,624,797

- **Panel A: 24-hour mean temperature**
  - 1°C increase $\rightarrow$ 2.8% increase in DV reports
  - Estimate aligns with semi-parametric results
- **Panel B: Max vs. Min temperature**
  - Both significant, but weaker than 24-hour mean
  - Literature often uses T_{max}, but may understate sustained exposure
- **Panel C: Daytime vs. Nighttime**
  - Daytime effect stronger; Nighttime smaller
  - Suggests exposure during daytime social/household interactions drives DV